

Pre RMO(INDIA) – 2019 (SET-1)

Instructions:

- A. Questions 1 to 6 carry 2 marks each; questions 7 to 21 carry 3 marks each; questions 22 to 30 carry 5 marks each.
 - B. All questions are compulsory.
 - C. There are no negative marks.
1. From a square with sides of length 5, triangular pieces from the four corners are removed to form a regular octagon. Find the area removed to the nearest integer?
 2. Let $f(x) = x^2 + ax + b$. If for all nonzero x , $f\left(x + \frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and the roots of $f(x) = 0$ are integers, what is the value of $a^2 + b^2$?
 3. Let x_1 be a positive real number and for every integer $n \geq 1$ let $x_{n+1} = 1 + x_1 x_2 \dots x_{n-1} x_n$. If $x_5 = 43$, what is the sum of digits of the largest prime factor of x_6 ?
 4. An ant leaves the anthill for its morning exercise. It walked 4 feet east and then makes a 160° turn to the right and walks 4 more feet. It then makes another 160° turn to the right and walks 4 more feet. If the ant continues this pattern until it reaches the anthill again. What is the distance in feet it would have walked?
 5. Five persons wearing badges with numbers 1, 2, 3, 4, 5 are seated on 5 chairs around a circular table. In how many way can they be seated so that no two persons whose badges have consecutive numbers are seated next to each other? (two arrangements obtained by rotation around table are considered different.)
 6. Let $\overline{abc}$ be a three digit number with nonzero digits such that $a^2 + b^2 = c^2$. What is the largest possible prime factor of $\overline{abc}$?
 7. On a clock, there are two instants between 12 noon and 1 PM, when the hour hand and the minute hand are at right angles. The difference in minutes between these two instants is written as $a + \frac{b}{c}$, where a, b, c are positive integers, with $b < c$ and b/c in the reduced form. What is the value of $a+b+c$?
 8. How many positive integers n are there such that $3 \leq n \leq 100$ and $x^{2n} + x + 1$ is divisible by $x^2 + x + 1$?
 9. Let the rational number p/q be closest to but not equal to $22/7$ among all rational numbers with denominator < 100 . What is the value of $p - 3q$?
 10. Let ABC be a triangle and Ω be its circumcircle. The internal bisectors of angles A, B and C intersect Ω at A_1, B_1 and C_1 respectively and the internal bisectors of angles A_1, B_1 and C_1 of the triangle $A_1 B_1 C_1$ intersect Ω at A_2, B_2 and C_2 respectively. If the smallest angle of triangle ABC is 40° , what is the magnitude of the smallest angle of the triangle $A_2 B_2 C_2$ in degrees?
 11. How many distinct triangles ABC are there, upto similarity, such that the magnitudes of angles A, B and C in degrees are positive integers and satisfy $\cos A \cos B + \sin A \sin B \sin C = 1$ for some positive integer k , where kC does not exceed 360° ?

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12. A natural number $k > 1$ is called good if there exist natural numbers $a_1 < a_2 < \dots < a_k$ such that $\frac{1}{\sqrt{a_1}} + \frac{1}{\sqrt{a_2}} + \dots + \frac{1}{\sqrt{a_k}} = 1$. Let $f(n)$ be the sum of the first n good numbers, $n \geq 1$. Find the sum of all values of n for which $\frac{f(n+5)}{f(n)}$ is an integer.
13. Each of the numbers $x_1, x_2, \dots, x_{101}$ is ± 1 . What is the smallest positive value of $\sum_{1 \leq i < j \leq 101} x_i x_j$?
14. Find the smallest positive integer $n \geq 10$ such that $n+6$ is a prime and $9n+7$ is a perfect square.
15. In how many ways can a pair of parallel diagonals or a regular polygon of 10 sides be selected?
16. A pen costs Rs. 13 and a notebook costs Rs. 17. A school spends exactly Rs. 10000 in the year 2017-18 to buy x pens and y notebooks such that x and y are close as possible (i.e. $|x - y|$ is minimum). Next year in 2018-19 the school spends little more than Rs. 10000 and buys y pens and x notebooks. How much more did the school pay?
17. Find the number of ordered triples (a, b, c) of positive integers $30a + 50b + 70c \leq 343$.
18. How many ordered pairs (a, b) of positive integers $a < b$ and $100 \leq a, b \leq 1000$ satisfy $\gcd(a, b) : \text{lcm}(a, b) = 1:495$?
19. Let AB be the diameter of a circle and let C be a point on the segment AB such that $AC:CB=6:7$. Let D be a point on the circle such that DC is perpendicular to AB . Let DE be the diameter through D . If $[XYZ]$ denotes the area of the triangle XYZ , find $[ABD]/[CDE]$ to the nearest integer.
20. Consider the set E of all natural numbers n such that when divided by 11, 12 and 13 respectively the remainders in that order are distinct primes number in arithmetic progression. If N is the largest number in E , find the sum of digits of N .
21. Consider the set $E = \{5, 6, 7, 8, 9\}$. For any partition $\{A, B\}$ of E , with both A and B non empty consider the number obtained by adding the product of elements of A to the product of elements of B . Let N be the largest prime number among these numbers. Find the sum of the digits of N .
22. What is the greatest integer not exceeding the sum $\sum_{n=1}^{1599} \frac{1}{\sqrt{n}}$?
23. Let $ABCD$ be a convex cyclic quadrilateral. Suppose P is a point in the plane of the quadrilateral such that sum of its distances from the vertices of $ABCD$ is the least. If $\{PA, PB, PC, PD\} = \{3, 4, 6, 8\}$, what is the maximum possible area of $ABCD$?
24. A $1 \times n$ rectangle ($n \geq 1$) is divided into n units of (1×1) squares. Each square of this rectangle is coloured red, blue or green. Let $f(n)$ be the number of colouring of the rectangle in which there are an even number of red squares. What is the largest prime factor of $f(9)/f(3)$? (the number of red squares can be zero)
25. A village has a circular wall around it, and the wall has four gates pointing north, south, east, west. A tree stands outside the village, 16 m north of the north gate and it can be just seen appearing on the horizon from a point 48 m east of the south gate. What is the diameter, in metres of the wall that surrounds the village?
26. Positive integers x, y, z satisfy $xy + z = 160$. Compute the smallest possible value of $x + yz$.
27. We will say that a rearrangement of the letters of a word has no fixed letters if, when the arrangement is placed below the word, no column has the same letters repeated. For instance HBRATA is a rearrangement with no fixed letter of BHARAT. How many distinguishable rearrangement with no fixed letters does BHARAT have? (the two A's are considered identical)

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28. Let ABC be the triangle with sides 51, 52, 53. Let Ω denote the incircle of triangle ABC . Draw tangents to Ω which are parallel to the sides of ABC . Let r_1, r_2, r_3 be the inradii of the three corner triangles so formed. Find the largest integer that does not exceed $r_1 + r_2 + r_3$.
29. In a triangle ABC , the median AD (with D on BC) and the angle bisector BE (with E on AC) are perpendicular to each other. If $AD = 7$ and $BE = 9$, find the integer nearest to the area of triangle ABC .
30. Let E denote the set of all natural numbers n such that $3 < n < 100$ and the set $\{1, 2, 3, \dots, n\}$ can be partitioned into 3 subsets with equal sums. Find the number of elements of E .

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